The Limits of Discrete-Event Schedulers in Interplanetary Network Simulation

Why Deep Space Breaks Conventional DES Paradigms

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International Conference on ns-3 (ICNS3 - 2026) · Lightning Talk

Focus: Continuous Astrodynamics vs. Discrete-Event Network Synchronization | **Mode:** Virtual Presentation

Deep Space Networks Are Not Like Earth Networks

How Terrestrial Simulators Work (e.g., ns-3)

- **Instant Signals:** On Earth, radio waves travel almost instantly (a few milliseconds across continents).
- **Event-Driven Clock:** The computer simulator sleeps until a packet is sent or received (an "event").
- **Frozen Moments:** It assumes nodes stay put or move predictably during message exchange.
- **Great for Earth:** Perfect for Wi-Fi, 5G, and low-Earth satellites.

Why Deep Space Breaks Assumptions

- **Extreme Delays:** Light takes **up to 24 minutes** to reach Mars, and over 1.3 seconds to reach the Moon.
- **Planets Never Stop:** Spacecraft and planets hurtle through continuous gravitational paths at tens of thousands of km/h.
- **Moving Target Problem:** While a signal travels, the receiver moves thousands of kilometers.
- **Clocks Run Differently:** Strong gravity near planets makes space clocks drift relative to Earth.

The Reality Check: In deep space, network connectivity is governed by continuous planetary motion, not discrete computer events.

The Dilemma: Two Flawed Traditional Approaches

Approach A: Rigid Fixed Ticks (e.g., Every 60s)

The simulator checks positions on a rigid clock:

- **Missed Connections:** If a spacecraft emerges from behind a planet between ticks, the simulator notices up to **30 seconds late**.
- **False Outages:** Drops packets and triggers false retransmission timeouts.
- **Massive Inefficiency:** Wastes thousands of redundant checks during hours of empty, quiet space cruise.

Approach B: Pure Event-Driven (Only on Packets)

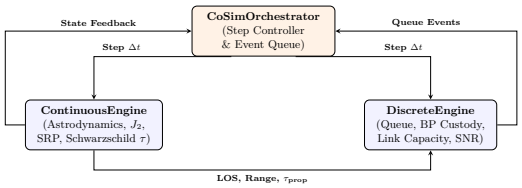
The simulator only updates when data packets move:

- **Frozen Physics:** During hours or days of communication blackouts, planetary motion is frozen in place!
- **Ghost Positions:** When the next packet arrives, the spacecraft position is wrong by **over 30,000 kilometers**!
- **Corrupted Link Budgets:** Completely miscalculates antenna signal strength.

The Incompatibility Rule: You cannot achieve physical flight accuracy, pinpoint contact timing, and fast execution with rigid, old simulation rules.

The Solution: Adaptive Multi-Rate Co-Simulation

Two Engines Working in Harmony



Flight Engine (Continuous) $\longleftrightarrow$ Network Engine (ns-3)

Smart Step Controller: Zoom In & Zoom Out

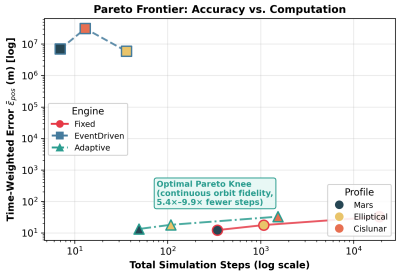
- **In Quiet Cruise:** Takes large strides in time (up to 1 hour) to run simulations at blazing speed.
- **Near Critical Events:** Automatically zooms in to split-second precision right as a satellite enters or leaves planetary shadow.
- **Zero Boundary Delay:** Catches every contact start and end with **0.0 ms error** without burning CPU time.

True Time-Weighted Accuracy

Measures real orbital error across continuous mission time rather than artificially favoring fixed snapshots.

Benchmark Proof: Faster, Lighter, and Exact

Accuracy vs. Compute Effort



Optimal Knee: 5x to 10x fewer computation steps with zero loss in orbital fidelity.

Tested on 3 Realistic Mission Profiles

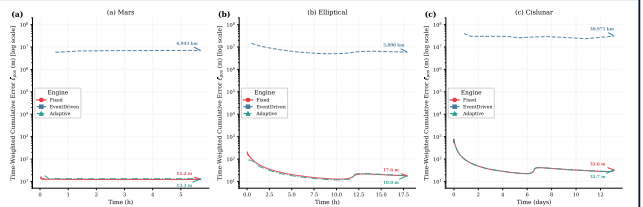
- **Mars Occultation:** Passing behind the red planet.
  - Runs **5.4x faster** than fixed stepping.
  - **0.0 ms timing error** (vs. 30 seconds late in fixed).
- **Molniya Orbit:** High-speed flyby near Earth.
  - Runs **8.1x faster**; trajectory stays within 18 meters.
- **13-Day Moon NRHO:** Long lunar cruise.
  - Runs **9.9x faster**, saving over 17,000 wasted steps.

Scalability (10 to 500 Spacecraft)

Simulates 500 space nodes in just 6.35 seconds with zero software lockups.

The Ghost Satellite Problem: Drifting into Nowhere

Position Drift Over Long Missions



Pure event-driven DES explodes in error during radio silence; our adaptive engine stays flat on the real flight path.

The Radio Silence Trap

- When no radio packets are sent, traditional event simulators go completely blind.
- In our 13-day lunar mission, the simulated satellite drifts by **over 30,970 kilometers**!
- That is more than **double the diameter of Earth**!

Our Adaptive Accuracy

- Our engine continuously tracks real orbital gravity in the background.
- Maximum position drift is bounded to **meters**, perfectly matching NASA SPICE standards.

What Goes Wrong When Simulators Lie?

1. False Connection Predictions

- A 30,000 km position error alters calculated radio signal strength by **up to 14 dB**.
- The simulator falsely reports a healthy high-speed link when the spacecraft is actually in **total radio blackout**.

2. Phantom Data Speeds

- Modern radios adapt speed to signal quality.
- Stale positions cause the simulator to schedule peak data transmission (20 Mbps) when the actual channel is dead (0 Mbps).

3. Panic Retransmission Storms

- Deep space network timers rely on expected travel delay.
- Underestimating delay makes timers expire prematurely, triggering storms of duplicate packets that choke the network.

4. Catastrophic Lost Science Data

- Because the simulator claims data is flowing safely, mission planners do not realize onboard storage is overflowing.
- Precious scientific photos and telemetry are permanently deleted.

The Golden Rule: *In deep space, a flight mechanics tracking error directly becomes a catastrophic networking failure.*

Three Practical Blueprints for ns-3

Blueprint 1: Aim Where the Spacecraft Will Be (In-Flight Light Solver)

- **Today's Limitation:** ns-3 calculates signal delay based on where the spacecraft was when the transmission started.
- **Our Proposal:** Teach ns-3 to solve where the moving target will actually be when the signal arrives minutes or hours later.

Blueprint 2: Smart Early-Warning Alarms (Boundary Event Guarding)

- **Today's Limitation:** Researchers force the computer to take millions of tiny steps just to watch for when a satellite dips behind a planet.
- **Our Proposal:** Introduce smart look-ahead triggers so the simulator speeds through quiet space and only slows down at the exact threshold crossings.

Blueprint 3: Deep-Space Multi-Clock Synchronization

- **Today's Limitation:** ns-3 uses a single global clock that assumes time ticks identically everywhere.
- **Our Proposal:** Support local spacecraft clocks that adjust for Einstein's gravitational and speed time dilation near massive planets.

Key Takeaways & Open-Source Code

What We Learned

- Deep space breaks traditional network simulators: multi-minute delays and continuous flight cannot be treated as simple frozen events.
- Our adaptive engine runs **5x to 10x faster** with **zero timing error** (0.0 ms) and sublinear scaling up to 500 nodes.
- Ready-to-implement blueprints to make ns-3 ready for the next era of interplanetary exploration.

Full Manuscript Under Peer Review

"The Limits of Discrete-Event Synchronization in Interplanetary Network Simulation: A Critical Review of Co-Simulation Fidelity"
Submitted to **Simulation Modelling Practice and Theory (Elsevier)**.

Scan for Code & Paper

github.com/PandiaJason/SPICE-ns-Project

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